1. OBJECTIVE

The objective of this project is to design and implement an ADS-B aircraft-tracking system that converts structured PingStation receiver data into useful aircraft information. Incoming surveillance records may be partial, repeated or stale, so the software decodes, validates, normalizes, stores and displays the data through a browser-based interface.

- Decode ICAO address, callsign, latitude, longitude, altitude, heading and speed when available.
- Normalize values into consistent units and reject unusable location records.
- Display current aircraft, route traces, status information and selected-aircraft details.
- Maintain local trajectory history for analysis and short-outage prediction.
- Export aircraft and trajectory data to CSV and Excel-compatible files.

2. REQUIREMENT ANALYSIS (SRS)

2.1 Introduction

Automatic Dependent Surveillance-Broadcast (ADS-B) enables equipped aircraft to broadcast identity, position and motion information for surveillance use [1], [2]. This project receives the higher-level stream supplied by a PingStation receiver, or replays a captured stream through the same decoder. It is an academic monitoring and data-logging application, not a replacement for certified air-traffic-control systems.

2.2 Scope

- Captured PingStation log replay and optional live TCP-stream reception.
- Frame separation, aircraft-field extraction and state merging.
- Validation, normalization, local storage, visualization and export.
- Short-gap trajectory extrapolation using recent observations; long-horizon flight-plan prediction is outside scope.